



# AERMOD Implementation Guide



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## AERMOD Implementation Guide

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Office of State Air Partnerships  
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## **Preface**

This document provides information on the recommended use of AERMOD to address specific issues and concerns related to the implementation of AERMOD for regulatory applications. The following recommendations augment the use of experience and judgment in the proper application of dispersion models. Advanced coordination with reviewing authorities, including the development of modeling protocols, is recommended for regulatory applications of AERMOD.



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## 1.0 What's new in this document? (7/2026, v26135)

### **Revisions Related to AERMOD Version 26135 (May 15, 2026):**

- Section 3.1.3 Use of AERSURFACE for determining surface characteristics

Updated discussion of the availability of NLCD data and added a discussion of the wind sector refinements for identifying a sector as either HIGHZ0 or LOWZ0 using the keyword SECTRING.

- Section 3.2.1 Upper air substitution

Added for use of the upper air substitution option added in AERMOD version 26135 and discusses the ability to now use prognostic upper air data with NWS surface observations in AERMET.

- Section 7.0 NO<sub>2</sub> Modeling

Added to note the differences in how the NO<sub>x</sub>, NO<sub>2</sub>, and O<sub>3</sub> chemistry are treated in the more recent GRSM NO<sub>2</sub> Tier 3 conversion versus the other Tier 3 options, PVMRM and OLM. Also discusses how source groups and background concentrations should be configured when using GRSM so the chemistry will be handled appropriately.

- Section 8.0 References

Moved from Section 7.0.

**Note:** Refer to the document [\*AERMOD System Bugs, Errata, and Related Guidance\*](#), available on the [\*EPA Support Center for Regulatory Atmospheric Modeling \(SCRAM\)\*](#) website, for supplemental guidance on program bugs/issues that are reported between version releases.

## **2.0 Document background and purpose**

### **2.1 Background (7/2026, v26135)**

In April 2005, the AERMOD Implementation Workgroup (AIWG) was formed in anticipation of AERMOD's promulgation as a replacement for the Industrial Source Complex (ISCST3) model. AERMOD fully replaced ISCST3 as the regulatory model on December 9, 2006 (EPA, 2005a), after a one-year grandfather period. The primary purpose for forming the AIWG was to develop a comprehensive approach to address implementation issues for which guidance is needed. The result of this initial AIWG was the publication of the first version of the AERMOD Implementation Guide on September 27, 2005.

In 2007, a new AIWG was formed as a standing workgroup to provide support to EPA's Office of Air Quality Planning and Standards (OAQPS) that is now the Office of Air State Partnerships (OSAP) after a reorganization of EPA in 2025. This document represents the combined efforts of the former AIWGs and OAQPS and is currently maintained by the Air Quality Modeling Branch (AQMB) in OSAP.

### **2.2 Purpose (10/2007)**

This document provides information on the recommended use of AERMOD to address a range of issues and types of applications. Topics are organized based on implementation issues, with additional information as appropriate on whether they impact the modules of the AERMOD modeling system (AERMOD, AERMET, and AERMAP), or related programs (AERSURFACE, AERSCREEN, AERMINUTE, and BPIPPRM). This document contains a section which highlights changes from the previous version, located in Section 1.0 of the document for use as a quick reference. Each section is also identified with month and year (mm/yyyy) representative of when the section was updated. More recent updates and additions include the corresponding model version number. Only sections with substantive changes or new recommendations are identified with new revision dates. Revision dates are not updated for sections with only minor edits to clarify the wording, correct typographical errors, or update citations to reference the most recent version of a document.

The recommendations contained within this document represent the current best use practices as determined by EPA. This document is designed to provide consistent, technically

sound recommendations to address specific issues and concerns relevant to the regulatory application of AERMOD. As always, advance coordination with the reviewing authorities on the application of AERMOD is advisable. Modeling protocols should be developed, and agreed upon by all parties, in advance of any modeling activity.

### 3.0 Meteorological data and processing

#### 3.1 Determining surface characteristics (01/2008)

When applying the AERMET meteorological processor (EPA, 2026b) to prepare the meteorological data for the AERMOD model (EPA, 2026d), the user must determine appropriate values for three surface characteristics: surface roughness length  $\{z_o\}$ , albedo  $\{r\}$ , and Bowen ratio  $\{B_o\}$ . The surface roughness length is related to the height of obstacles to the wind flow and is, in principle, the height at which the mean horizontal wind speed is zero based on a logarithmic profile. The surface roughness length influences the surface shear stress and is an important factor in determining the magnitude of mechanical turbulence and the stability of the boundary layer. The albedo is the fraction of total incident solar radiation reflected by the surface back to space without absorption. The daytime Bowen ratio, an indicator of surface moisture, is the ratio of sensible heat flux to latent heat flux and is used for determining planetary boundary layer parameters for convective conditions driven by the surface sensible heat flux. This section provides recommendations regarding several issues associated with determining appropriate surface characteristics for AERMOD modeling applications.

##### 3.1.1 Meteorological data representativeness considerations (01/09/2008)

When using National Weather Service (NWS) data for AERMOD, data representativeness can be thought of in terms of constructing realistic planetary boundary layer (PBL) similarity profiles and adequately characterizing the dispersive capacity of the atmosphere. As such, the determination of representativeness should include a comparison of the surface characteristics (*i.e.*,  $z_o$ ,  $B_o$ , and  $r$ ) between the NWS measurement site and the source location, coupled with a determination of the importance of those differences relative to predicted concentrations. Site-specific meteorological data are assumed to be representative of the application site; however, the determination of representativeness of site-specific data for AERMOD applications should also include an assessment of surface characteristics of the measurement and source locations and cannot be based solely on proximity. The recommendations presented in this section for determining surface characteristics for AERMET apply to both site-specific and non-site-specific (*e.g.*, NWS) meteorological data.

The degree to which predicted pollutant concentrations are influenced by surface parameter differences, between the application site and the meteorological measurement site, depends on the nature of the application (*i.e.*, release height, plume buoyancy, terrain influences, downwash considerations, design metric, etc.). For example, a difference in  $z_o$  for one application may translate into an unacceptable difference in the design concentration. For a different application the same difference in  $z_o$  may lead to an insignificant difference in design concentration. If the reviewing agency is uncertain as to the representativeness of a meteorological measurement site, a site-specific sensitivity analysis may be needed to quantify, in terms of expected changes in the design concentration, the significance of the differences in each of the surface characteristics.

If the proposed meteorological measurement site's surface characteristics are determined to NOT be representative of the application site, it may be possible that another nearby meteorological measurement site may be representative of both meteorological parameters and surface characteristics. If that is not the case, it is likely that site-specific meteorological data will be required.

### 3.1.2 Methods for determining surface characteristics (04/2021)

The availability of high-resolution digitized land cover databases provides an opportunity to apply systematic procedures to determine surface characteristics based on an objective analysis of the gridded land cover data across a domain. A proper analysis of such data must take into consideration the relationship between surface characteristics and the meteorological measurements on which the surface characteristics will be applied.

Based on model formulations and model sensitivities, the relationship between the surface roughness upwind of the measurement site and the measured wind speeds is generally the most important consideration. The effective surface roughness length should be based on an upwind distance that captures the net influence of surface roughness elements on the measured wind speeds needed to properly characterize the magnitude of mechanical turbulence in the approach flow. Studies have examined the response of the atmosphere to abrupt changes in the surface roughness and provide some insight into the relationship between measured winds and surface roughness [*e.g.*, Blom and Warenta (1969), Businger (1986), Högström and Högström (1978), Horst and Weil (1994), Irwin (1978), Rao, et al. (1974), and Taylor (1969)]. Such

changes in surface roughness result in the development of an internal boundary layer (IBL) which grows with distance downwind of the roughness change, and defines the layer influenced by the transition in surface roughness. The size and structure of the IBL is very complex, even for idealized cases of uniform roughness upwind and downwind of the transition. The IBL is also affected by the magnitude and direction of the roughness change and the stability of the upstream flow. The IBL generally grows more slowly for stable conditions than for neutral or unstable approach flow and will also tend to grow more slowly for rough-to-smooth transitions than for smooth-to-rough transitions. The relationship between surface roughness and measured wind speeds is even more complex in real world applications given the typically patchy nature of the heterogeneity of surface roughness elements.

The recommended upwind distance for surface roughness should account for the fact that surface roughness effects in AERMOD are more important for stable atmospheric conditions than for neutral/unstable conditions, and that meteorological monitoring sites are typically characterized by open (low roughness) exposures to accommodate recommended siting criteria (EPA, 2000). For typical measurement programs, including NWS stations, the reference wind measurements will be taken for an anemometer height of approximately 10 meters above ground. An upwind distance based on the recommended siting criterion of at least 10 times the height of nearby obstacles (EPA, 2000), which would correspond to about 100 meters for typical obstacles such as trees and 2-3 story buildings, is considered inadequate for this purpose. However, the previous recommendation to use an upwind distance of 3 kilometers for surface roughness is considered too large because the boundary layer up to typical measurement heights of 10 meters will generally respond to changes in roughness length over much shorter distances. Including land cover information across an upwind distance that is too large could misrepresent the amount of mechanical turbulence present in the approach flow and bias model results, especially for low-level releases.

The recommended upwind distance for processing land cover data to determine the effective surface roughness for input to AERMET is 1 kilometer relative to the location of the meteorological tower. This recommended distance is considered a reasonable balance of the complex factors cited in the discussion above. If land cover varies significantly by direction, then surface roughness should be determined based on sector. However, the width of the sectors

should be no smaller than a 30-degree arc. Further information on the definition of sectors for surface roughness is provided in the User's Guide for the AERMOD Meteorological Preprocessor (AERMET) (AERMET User's Guide) (EPA, 2026b). Exceptions to the recommended default distance of 1 kilometer for surface roughness may be considered on a case-by-case basis for applications involving site-specific wind speed measurements taken at heights well above 10 meters, in situations with significant discontinuities in land cover just beyond the recommended 1-kilometer upwind distance, or for sites with significant terrain discontinuities (*e.g.*, the top of a mesa or a narrow, steep valley). Another factor that may need to be considered in some cases for determining an effective surface roughness length, is the potential contribution of nearby terrain or other significant surface expression, not reflected in the land cover data, to the generation of mechanical turbulence. Use of a non-default distance for surface roughness estimation, or modification of surface roughness estimates to account for terrain/surface-expression effects, should be documented and justified in a modeling protocol submitted to the appropriate reviewing authority prior to conducting the modeling analysis.

The dependence of meteorological measurements and plume dispersion on Bowen ratio and albedo is very different than the dependence on surface roughness. Effective values for Bowen ratio and albedo are used to estimate the strength of convective turbulence during unstable conditions by determining how much of the incoming radiation is converted to sensible heat flux. These estimates of convective turbulence are not linked as directly with tower measurements as the linkage between the measured wind speed and the estimation of mechanical turbulence intensities driven by surface roughness elements. While local surface characteristics immediately upwind of the measurement site are very important for surface roughness, effective values of Bowen ratio and albedo determined over a larger domain are more appropriate.

The recommended approach for processing digitized land cover data to determine the effective Bowen ratio and albedo for input to AERMET is to average the surface characteristics across a representative domain without any direction or distance dependency. The recommended default domain is a 10 x 10-kilometer region centered on the measurement site. Use of the measurement location to define the domain is likely to be adequate for most applications. However, a domain representative of the application site may be more appropriate for some applications, particularly if most of the sources are elevated releases. The use of an alternative domain for Bowen ratio and albedo should be documented and justified in a modeling protocol

submitted to the appropriate reviewing authority prior to conducting the modeling analysis.

Beyond defining the appropriate domains to use for processing digitized land cover data, additional considerations are needed regarding the computational methods for processing of the data. Since the width of a sector increases with distance from the measurement site, the land cover farther from the site would receive a higher effective weight than land cover closest to the site if a direct area-weighted averaging approach were used to calculate an effective surface roughness. An inverse-distance weighting is recommended for determining surface roughness from digitized land cover data to adjust for this factor, since the length of an arc (across a sector) is proportional to the distance from the center. In addition, a geometric mean is recommended for calculating the effective surface roughness since the AERMOD formulations are dependent on the  $\ln(z_o)$ . Note that the arithmetic average of the  $\ln(z_o)$  is mathematically equivalent to the geometric mean of  $z_o$ . Since the Bowen ratio represents the ratio between sensible heat flux and latent heat flux, the use of a geometric mean is also recommended for calculating effective values of Bowen ratio. Geometric means are more appropriate for calculating “average” values of ratios; for example, the “average” for Bowen ratios of 0.5 and 2.0 should be 1.0, which is accomplished with the use of a geometric mean. A simple arithmetic average is recommended for calculating effective values of albedo.

These recommendations for determining surface characteristics supersede previous recommendations and should be followed unless case-by-case justification can be provided for an alternative method. The recommendations described above are briefly summarized below:

1. The determination of the **surface roughness length** should be based on an inverse-distance weighted geometric mean for a default upwind distance of 1 kilometer relative to the measurement site. Surface roughness length may be varied by sector to account for variations in land cover near the measurement site; however, the sector widths should be no smaller than 30 degrees.
2. The determination of the **Bowen ratio** should be based on a simple unweighted geometric mean (*i.e.*, no direction or distance dependency) for a representative domain, with a default domain defined by a 10 x 10 km region centered on the measurement site.
3. The determination of the **albedo** should be based on a simple unweighted arithmetic mean (*i.e.*, no direction or distance dependency) for the same representative domain as



defined for Bowen ratio, with a default domain defined by a 10 x 10 km region centered on the measurement site.

An important aspect of determining surface characteristics from digitized land cover data is the assignment of surface characteristic values for each of the parameters (surface roughness, Bowen ratio and albedo) to the land cover categories contained in the dataset. Several references are available to guide those assignments, including the User's Guide for AERSURFACE Tool (AERSURFACE User's Guide) (EPA, 2026c), the AERMET User's Guide (EPA, 2026b), Garrett (1992), Gifford (1968), Oke (1978), Randerson (1984), and Stull (1988). Due to the somewhat subjective nature of this process, and the fact that specific land cover categories may include a wide range of values for some surface characteristics, the methods and assumptions used to assign surface characteristics based on land cover categories should be thoroughly documented and justified.

### 3.1.3 Use of AERSURFACE for determining surface characteristics (7/2026, v26135)

EPA has developed a tool called AERSURFACE (EPA, 2026c) that can be used in determining reproducible surface characteristic values, including albedo, Bowen ratio, and surface roughness length, for input to AERMET. The current version of AERSURFACE, version 26135, supports the use of land cover data from the Multi-Resolution Land Characteristics (MRLC) Consortium National Land Cover Database (NLCD) representative land cover beginning with 1985 and forward. More information on the NLCD data products available from the MRLC and downloading instructions can be found in the "NLCD Sources for AERSURFACE" document available on SCRAM.<sup>1</sup> The NLCD provides land cover data at a spatial resolution of 30 meters. In 2024, new annual NLCD products were released for the contiguous U.S. Coverage of the NLCD for Alaska, Hawaii, and Puerto Rico is limited to select years of legacy NLCD. The NLCD for all years is now based on 16 categories with an additional four categories that are specific only to Alaska, for a total of 20 distinct categories. This is the same classification scheme used by the post-1992 NLCD years prior to the release of the annual NLCD in 2024. The legacy 1992 classification scheme is no longer in use and the ability to process the legacy 1992 NLCD files has been removed from AERSURFACE beginning with

<sup>1</sup>[https://gaftp.epa.gov/Air/aqmg/SCRAM/models/related/aersurface/NLCD\\_Sources\\_for\\_AERSURFACE.pdf](https://gaftp.epa.gov/Air/aqmg/SCRAM/models/related/aersurface/NLCD_Sources_for_AERSURFACE.pdf)

version 26135.

Further, the NLCD includes separate data files of representative percent impervious and percent canopy beginning with year 1985 which can be input to AERSURFACE to supplement the land cover data. EPA recommends supplementing land cover data with both percent impervious and percent canopy data when both are available for the same NLCD year concurrent with the land cover data. In cases where only one is available or neither is available, EPA recommends processing the land cover data without supplementation of percent impervious or percent canopy data.

AERSURFACE incorporates look-up tables of representative surface characteristic values by land cover category and seasonal category. Further details regarding application of the AERSURFACE tool are provided in the AERSURFACE User's Guide (EPA, 2026c). The AERSURFACE tool incorporates the recommended methods for determining surface characteristics from digitized land cover data described in Section 3.1.2. While the AERSURFACE tool is not currently considered to be part of the AERMOD regulatory modeling system, the recommended methodology described in Section 3.1.2 should be followed unless case-by-case justification can be provided for an alternative method. The methods described in the AERSURFACE User's Guide for supplementing land cover data with percent impervious and percent canopy where both data products are available, should also be followed.

Beginning with AERSURFACE version 20060, wind sectors used for determining surface roughness length around the meteorological tower can be specified individually as an airport (AP) or non-airport (NONAP) sector based on whether areas within a sector categorized as developed (21-24) consists of large impervious areas represented by runways with grassy areas in between or the impervious areas are primarily buildings or other structures. Impervious areas in sectors specified as airport sectors are more heavily weighted toward lower roughness length values than those that are specified as non-airport sectors.

With the release of AERSURFACE version 24142, new keywords were added to extend this weighting of higher and lower roughness for developed categories to the area around site-specific meteorological towers that are not located at airports. These new keywords include LOWZ0 and HIGHZ0, and can be used interchangeably with AP and NONAP, respectively, to indicate whether a sector should be weighted toward a higher or lower roughness value based on

the land use within areas characterized as developed, regardless if the area is at or near an airport. Similarly, the VARYZ0 keyword (analogous to the VARYAP keyword) has been added to indicate that wind sectors will be assigned individually.

Beginning with AERSURFACE version 26135, a refinement has been added when assigning LOWZ0 and HIGHZ0 to surface roughness wind sectors. To further delineate the land use within a wind sector that is primarily defined as “Developed” (categories 21-24) or as “Cultivated Crops” (category 82), users now have the option to divide sectors into annular rings (200 m wide, radially) and assign LOWZ0 or HIGHZ0 to the individual ring segments within a sector using SECTRING flag with FREQ\_SECT keyword, rather than defining the entire sector as either LOWZ0 or HIGHZ0. AERSURFACE calculates the average surface roughness length in the same manner as the previous version, 24142, but the land cover value assigned to a specific 30m x 30m pixel used in calculating the average is based on the LOWZ0 or HIGHZ0 flag assigned to the sector and ring segment the pixel is located. A ring segment largely comprised of “Developed” land cover that is primarily buildings and/or trees would be assigned HIGHZ0 while a ring segment comprised mostly of runways and/or ground-level parking lots would be assigned LOWZ0. Consult the AERSURFACE User’s Guide (EPA, 2026c) for additional information on defining wind sectors for determining surface roughness length.

### **3.2 Upper air data (11/2024, v24142)**

The AERMET meteorological processor requires full upper air soundings (radiosonde data) representing the vertical potential temperature profile near sunrise, to calculate convective mixing heights. For AERMOD applications within the U.S., the early morning sounding, nominally collected at 12Z (or UTC/GMT), is typically used for this purpose. The formats that AERMET can process are TD-6201, FSL, and IGRA (Integrated Global Radiosonde Archive). Prior to the autumn of 2024, historical and current upper air data in the FSL format were available for download, free of charge, from the online NOAA/ESRL Radiosonde Database (<https://ruc.noaa.gov/raobs/>). However, beginning in the autumn of 2024, historical and current upper air data in the FSL format are no longer available. For any applications that require a download of upper air data after the autumn of 2024, IGRA data will be the only available data. The IGRA data and specifications can be downloaded free of charge from the NCEI ftp server (<ftp://ftp.ncei.noaa.gov/pub/data/igra> and <https://www.ncei.noaa.gov/pub/data/igra/>).

Beginning with AERMET version 24142, the user is now required to enter the upper air station's elevation or AERMET will issue an error and abort processing. In previous versions of AERMET, the user did not have to enter the elevation. However, recent years of IGRA data have missing heights for the surface level of some soundings. The surface level height is needed by AERMET to convert the sounding levels to height above ground. If the surface level height is missing, AERMET will issue a warning and skip the sounding. If the skipped sounding is the early morning sounding used to calculate the convective mixing heights for the day, that day will have missing convective mixing heights and associated parameters ( $w^*$  and potential temperature lapse rate) and concentrations will not be calculated for those hours.

To alleviate this issue, AERMET 24142 has been modified to substitute the most recent non-missing surface level in the sounding file or station elevation entered via the LOCATION keyword in the UPPERAIR pathway for missing surface levels.

### 3.2.1 Upper air substitution (07/2026 v26135)

In recent years, the availability of upper air data to calculate convective mixing heights in AERMET has become increasingly problematic due to station closures and helium supply issues. This can result in gaps in upper air data, which leads to missing soundings and missing convective hours in AERMOD simulations, leading to unrepresentative concentrations. Beginning with AERMET 26135, users can now substitute missing soundings with soundings from days prior to or after a day with missing soundings, either from the same station or a secondary station. The substitution is initiated by the use of the UPP\_SUB action with the METHOD keyword in Stage 2 of AERMET with several options. The substitution hierarchy and keywords are explained in Section 3.7.7.8 of the AERMET User's Guide. Details on inputting a secondary station are in Section 3.4.1 of the AERMET User's Guide.

Also beginning with version 2135, users can now use prognostic upper air data with observed surface data when there are no representative upper air stations nearby the application site. There is no special processing or keywords in AERMET to use prognostic upper air data with observed surface data. Users just input the UPPERAIR data output by MMIF using the UPPERAIR pathway in AERMET. The data should contain the same data period as the surface data (five years for NWS surface data, at least one year of site-specific data) and should be for the location of the surface station (either NWS or site-specific tower). Users should refer to the

latest MMIF user's guide and guidance on running MMIF for AERMOD when processing prognostic data in MMIF for AERMOD.

### 3.3 Processing site-specific meteorological data for urban applications (04/2021)

The use of site-specific meteorological data obtained from an urban setting may require some special processing if the measurement site is located within the influence of the urban heat island and site-specific turbulence measurements are available (*e.g.*,  $\sigma_\theta$  and/or  $\sigma_w$ ). As discussed in Section 5.4, the urban algorithms in AERMOD are designed to enhance the turbulence levels relative to the nearby rural setting during nighttime stable conditions to account for the urban heat island effect. Since the site-specific turbulence measurements will reflect the enhanced turbulence associated with the heat island, site-specific turbulence measurements should not be used when applying AERMOD's urban option, in order to avoid double counting the effects of enhanced turbulence due to the urban heat island. If site-specific measurements are available for the site, the user can choose to not include them in AERMET or they can be processed in AERMET and the new turbulence options, NOTURB or NOTURBST can be used to ignore the turbulence measurements. The NOTURB option will reset non-missing turbulence to missing when reading the profile file for all hours and the NOTURBST option will reset the non-missing turbulence values to missing for stable hours only, when the enhanced turbulence due to the urban heat island is applied in the model. See Section 3.5.9 of the User's Guide for the AMS/EPA Regulatory Model (AERMOD) (AERMOD User's Guide) (EPA, 2026d) for more information about the options.

As also discussed in Section 5.4 of this document, the AERMOD urban option (URBANOPT) should be selected for urban applications, regardless of whether the meteorological measurement site is in an urban setting since the limited surface meteorological measurements available from the meteorological measurement program (even with measured turbulence) will not adequately account for the meteorological characteristics of the urban boundary layer included in the AERMOD urban algorithms.

### 3.4 Use of prognostic meteorological model data as inputs to AERMOD (04/2018)

In recent years, interest has grown in the use of prognostic meteorological data, such as the Weather Research and Forecasting (WRF) model to create inputs for dispersion modeling

with AERMOD. This is especially true in locations where it can be difficult to find an adequately representative NWS station or cost-prohibitive or not feasible to set up a site-specific meteorological monitoring tower. As part of the recent update to Appendix W to 40 CFR Part 51 (*Guideline*) (EPA,2024b), the use of prognostic data is allowed for regulatory applications of AERMOD where it is cost-prohibitive or not feasible to collect site-specific data and there is no representative NWD or comparable station nearby. EPA developed the Mesoscale Model Interface Program, or MMIF for processing prognostic meteorological data for AERMOD (Ramboll Environ, 2022). For more information see Section 8.4.5 of the *Guideline* (EPA, 2024b) and the Guidance on the Use of the Mesoscale Model Interface Program (MMIF) for AERMOD Applications (MMIF guidance document) (EPA, 2024a) that offers recommendations on the use of prognostic data and MMIF.

### **3.5 Use of COARE algorithms in AERMET for overwater applications (11/2024, v24142)**

In recent years, applications of AERMOD in a marine boundary layer have increased. These applications have involved the use of the AERCOARE pre-processor as an alternative model to process observed buoy data or overwater prognostic data to generate the meteorological inputs for AERMOD. AERCOARE is a pre-processor that invokes the Coupled Ocean-Atmosphere Response Experiment algorithms to calculate boundary layer parameters such as surface friction velocity and Monin-Obukhov length for AERMOD, rather than using AERMET to process the data. The algorithms from COARE are more representative of the marine boundary layer rather than the algorithms in AERMET which are more appropriate for land-based stations (i.e., NWS stations or meteorological towers) or land-based prognostic data. The use of AERCOARE with AERMOD has been considered an alternative model approach under the guise of the *Guideline*, because the Offshore Coastal Dispersion model (OCD) is the preferred model for overwater applications per Addendum A of the *Guideline*.

As AERMOD use in overwater applications has increased, the COARE algorithms were incorporated into AERMET/AERMOD version 23132 as a beta option as part of the 2023 proposed update to the *Guideline*. With version 24142 of AERMET and AERMOD, the COARE algorithms are now considered a regulatory non-default option and the use of AERCOARE is no longer needed to process overwater meteorological data for AERMOD. COARE was included in AERMET to eliminate the alternative model approval when using

AERMOD for overwater applications. This is because AERMET is the regulatory meteorological pre-processor for AERMOD per Addendum A in the *Guideline*. Also, inclusion of the COARE algorithms in AERMET ensure that the COARE algorithms will be routinely updated with routine AERMET updates. Note, the alternative model approval that is not needed for AERMOD applications overwater is contingent that shoreline fumigation and platform downwash are adequately addressed and there is consultation with the appropriate reviewing authority and Regional Office.

COARE is invoked in AERMET via the METPREP pathway in Stage 2 of AERMET using the METHOD keyword along with COARE and RUN\_COARE options. Additionally, there are recommended options for running COARE in AERMET, and these recommendations are listed in Section 3.14 of the AERMET User's Guide for either observed buoy data or prognostic overwater data. The user is urged to check the most recent AERMET User's Guide for current recommendations.

## 4.0 Terrain data and processing

### 4.1 Modeling sources with terrain-following plumes in sloping terrain (01/2008)

Under the regulatory default mode (DFAULT option on the MODELOPT keyword), for all situations in which there is a difference in elevation between the source and receptor, AERMOD simulates the total concentration as the weighted sum of two plume states (Cimorelli, *et al.*, 2004): 1) a horizontal plume state (where the plume's elevation is assumed to be determined by release height and plume rise effects only, and thereby allowing for impingement if terrain rises to the elevation of the plume); and 2) a terrain-responding plume state (where the plume is assumed to be entirely terrain following).

For cases in which receptor elevations are lower than the base elevation of the source (*i.e.*, receptors that are down-slope of the source), AERMOD will predict concentrations that are less than what would be estimated from an otherwise identical flat terrain situation. While this is appropriate and realistic in most cases, for cases of down-sloping terrain where expert judgment suggests that the plume is terrain-following (*e.g.*, down-slope gravity/drainage flow), AERMOD will tend to underestimate concentrations when terrain effects are taken into account. AERMOD may also tend to underestimate concentrations relative to flat terrain results for cases involving low-level, non-buoyant sources with up-sloping terrain since the horizontal plume component will pass below the receptor elevation. Sears (2003) has examined these situations for low-level area sources and has shown that as terrain slope increases the ratio of estimated concentrations from AERMOD to ISC (which assumes flat terrain for area sources) decreases substantially.

To avoid underestimating concentrations in such situations, it may be reasonable in cases of terrain-following plumes in sloping terrain to apply the non-DFAULT option to assume flat, level terrain. This determination should be made on a case-by-case basis, relying on the modeler's experience and knowledge of the surrounding terrain and other factors that affect the air flow in the study area, characteristics of the plume (release height and buoyancy), and other factors that may contribute to a terrain-following plume, especially under worst-case meteorological conditions associated with the source. The decision to use the non-DFAULT option for flat terrain, and details regarding how it will be applied within the overall modeling analysis, should be documented and justified in a modeling protocol submitted to the appropriate



reviewing authority prior to conducting the analysis.

## 4.2 AERMAP domain (09/2005)

Section 2.1.2 of the User's Guide for the AERMOD Terrain Preprocessor (AERMAP) (AERMAP User's Guide) (EPA, 2024c) states that the AERMAP domain must include all terrain features that exceed a 10% elevation slope from any given receptor. The 10% slope rule may lead to excessively large domains in areas with considerable terrain features (*e.g.*, fjords, successive mountain ranges, etc.). In these situations, the reviewing authority may make a case-by-case determination regarding the domain size needed for AERMAP to determine the critical dividing streamline height for each receptor.

## 4.3 Terrain elevation data sources for AERMAP (04/2021)

AERMAP was revised (beginning with version 09040) to support processing of terrain elevations from the National Elevation Dataset (NED) developed by the U.S. Geological Survey, which has transitioned to the USGS 3D Elevation Program (3DEP) (Sugarbaker, et al., 2017). More information about the USGS 3DEP can be found at <https://www.usgs.gov/3d-elevation-program>. 3DEP data can be downloaded using the USGS National Map downloader at <https://apps.nationalmap.gov/downloader/>. For the purposes of this guide, the terms NED and 3DEP will be used interchangeably.

AERMAP still supports terrain elevations in the older formats referenced as Digital Elevation Model (DEM) files (*e.g.*, 7.5-minute and 1-degree DEM files) and has also been enhanced to process DEM files of mixed format in the same run. AERMAP currently does not support processing of elevation data in both the DEM format and the GeoTIFF format for 3DEP data in the same run.

The USGS DEM archives are now static and will not be updated in the future, while the 3DEP data are being actively supported and checked for quality. Therefore, 3DEP represents a more up-to-date and improved resource for terrain elevations for use with AERMAP. Due to problems that have been encountered with DEM data, AERMOD users are encouraged to use 3DEP data. Problems encountered with DEM data include incorrect geo-referencing information for entire DEM files and elevations that reflect the tops of buildings and trees in some cases. The use of 3DEP data should avoid these issues and provide additional advantages over the use

of DEM data, including consistent horizontal resolution and reference datum (generally NAD83 or WGS84). Some applications of AERMAP using DEM data may involve inconsistent reference datums for adjacent DEM files, which can result in receptors being located within gaps between files due to the datum shift. Gaps may also occur within DEM files generated by various software tools to convert from one format to another when a NAD conversion is involved, *e.g.*, converting 1-degree DEM data to the 7.5-minute DEM format to fill areas not covered by available 7.5-minute data. The AERMAP User's Guide (EPA, 2024c) provides a more detailed discussion of issues associated with gaps between DEM files or within DEM files and describes how these cases are handled by AERMAP.

While 3DEP is considered an improvement in the quality and consistency of elevation data for use with AERMAP, there have been some issues associated with the GeoTIFF format supported by AERMAP of which users should be aware. One issue of importance to AERMAP users is that the 3DEP GeoTIFF files currently available from the USGS may not include the GeoKey specifying the units for the elevation data. The USGS documentation for NED data (now 3DEP) (USGS, 2002) indicates that elevations are in units of meters and are provided in floating point format. AERMAP will therefore assume units of meters if the elevation units GeoKey is absent. However, non-standard (*i.e.*, non-USGS) elevation data in GeoTIFF format may not be in units of meters. AERMAP provides an option for users to specify elevation units in these cases. However, users must exercise caution in using such data unless the correct units can be confirmed. The AERMAP User's Guide (EPA, 2024c) provides a more detailed discussion of these and other potential issues associated with the GeoTIFF format supported for 3DEP data.

A second issue is the elevation data downloaded from the USGS National Map download site in GeoTIFF format cannot be read directly by AERMAP and must be converted to remove the internal compression of the data. Instructions for removing the compression are provided at [https://gaftp.epa.gov/Air/aqmg/SCRAM/models/related/aermap/Access\\_and\\_Conversion\\_of\\_Elevation\\_Data\\_for\\_AERMAP.pdf](https://gaftp.epa.gov/Air/aqmg/SCRAM/models/related/aermap/Access_and_Conversion_of_Elevation_Data_for_AERMAP.pdf).

The 3DEP elevation data are currently available as GeoTIFF files for the conterminous United States, Hawaii, Puerto Rico, and the Virgin Islands at a horizontal resolution of 1 arc-second (approximately 30 meters), and at a resolution of 2 arc-seconds for Alaska. Higher

resolution data at 1/3<sup>rd</sup> arc-second (about 10 meters) are available for most areas outside of Alaska, and even 1/9<sup>th</sup> arc-second data (about 3 meters) are available for some areas but do not appear to be available for download as GeoTIFF files. The appropriate horizontal resolution for the input terrain data and receptor network should be determined in consultation with the reviewing authority based on the specific needs of the project. Higher resolutions for both the terrain data and receptor network may be necessary in areas with significant terrain relief than for areas with relatively flat terrain. While acceptable, using the highest resolution elevation data available for determining receptor elevations and hill height scales may not always be justified. Since spatial coverage of terrain data for some resolutions may not be complete, it is also worth noting that use of a single resolution across the domain has advantages, and AERMAP places some restrictions on the order of DEM or 3DEP file inputs when mixed resolution data are used.

Regardless of the receptor and terrain data resolutions used in AERMAP, it is advisable to check the accuracy of receptor elevations and hill height scales being input to AERMOD for significant terrain features that are likely to be associated with peak concentrations based on proximity and elevation in relation to the sources. Elevations for fence line or other nearby receptors located within areas that have been altered due to facility construction may require special consideration since these changes in local topography may not be reflected in the USGS terrain files. Use of receptor elevations derived from plant survey data may be an acceptable alternative in these cases. The option available in AERMAP for the user to provide elevations may be utilized to determine hill height scales for these special cases, rather than the default option of determining elevations and hill height scales based on the input terrain data. However, care should be exercised to ensure that the hill height scales determined by AERMAP are also representative of the modified topography. If alternative data sources and/or methods are used to estimate receptor elevations, users must recognize that receptor elevations input to AERMOD should represent the best estimate of the actual terrain elevation at the receptor location. Use of a “conservative” estimate of the maximum elevation in the vicinity of the receptor location, such as the maximum within a “grid cell” centered on the receptor, is not appropriate for use in AERMOD based on the formulation of the terrain algorithms in the model and may not result in a conservative estimate of concentrations.

Beginning with the version dated 09040, AERMAP can also process terrain elevations derived from the Shuttle Radar Topography Mission (SRTM) data, provided they are stored as

GeoTIFF files that are compatible with AERMAP. 3DEP data represent the ground (“bare earth”) elevation, which is a more appropriate input for determining receptor elevations and hill height scales for use in AERMOD. AERMOD users should therefore avoid the use of SRTM data to determine elevations for use in AERMOD. However, SRTM data are available for most of the globe, and may be the only practical alternative for applications beyond the U. S. While AERMAP can process both 3DEP and SRTM data in GeoTIFF format in the same run, the only situation that might warrant such an approach would be applications along a border that extends beyond the domain covered by the 3DEP data. The SRTM elevation data are typically based on the WGS84 horizontal datum, rather than the NAD83 datum used for most 3DEP data. While AERMAP treats the WGS84 and NAD83 datums as equivalent, AERMAP will issue a warning message for any terrain file input as NED data that is not in the NAD83 datum to flag the possibility that non-NED data (or non-standard NED data) are being used.

Given the number of options available for elevation data inputs to AERMAP, and the range of issues associated with elevation data, users are encouraged to clearly document the source of elevation data used for AERMOD applications in the modeling protocol, including the resolution and horizontal reference datum for the data and any pre-processing that might have been done, such as converting from one format to another. Since the 3DEP data are being checked for quality and updated as needed, AERMAP users should also consider acquiring updated terrain files on a periodic basis before use in regulatory modeling applications. If the option to provide receptor elevations to AERMAP is utilized, rather than using the default option of determining elevations based on the input terrain data, the sources and methods used to determine the provided elevations should be clearly documented along with a justification for use of that option.

As changes have occurred over time regarding availability of the elevation data and file formats provided by the USGS, refer to the AERMAP User’s Guide (EPA, 2024c) and the EPA SCRAM website (<https://www.epa.gov/scram>) for current information on data coverage, resolution, format, and access to elevation data. Additional information on NED/3DEP data products and specifications can be found at the USGS National Map website at <https://www.usgs.gov/3d-elevation-program>.

#### **4.4 Manually entering terrain elevations in AERMAP (03/2009)**

AERMAP currently does not have the capability of accepting hand-entered terrain data in an “xyz” format. AERMAP only accepts terrain data from digitized elevation files in the DEM or 3DEP/GeoTIFF formats. Therefore, if no DEM or 3DEP /GeoTIFF data are available for a particular application, terrain elevations may need to be determined through other means. One option may be to manually enter gridded terrain elevations in a form that mimics the DEM data format. Instructions for how to accomplish this can be found on the SCRAM web site <https://www.epa.gov/scram> in a document titled “On inputting XYZ data into AERMAP.”

As noted in Section 4.3, if alternative sources and/or methods are used to estimate receptor elevations, users must recognize that receptor elevations input to AERMOD should represent the best estimate of the actual terrain elevation at the receptor location, and these alternative sources and methods should be documented in the modeling protocol. As also noted in Section 4.3, SRTM elevation data in GeoTIFF format is available for most of the globe, which may provide another alternative source of elevation data for use in AERMAP. However, SRTM data represents the heights of obstacles, such as buildings and trees, rather than ground elevations, and should be used with caution and only as a last resort.

#### **4.5 Use of AERMAP to determine source elevations (03/2009)**

AERMAP includes the capability of estimating terrain elevations for sources based on the same data and procedures used to estimate receptor elevations. However, the requirements for determining source elevations are somewhat different than the requirements for determining receptor elevations since a greater emphasis is placed on the accuracy of elevations at specific locations in the case of sources. While the accuracy of specific receptor elevations is also important, the main focus for receptors should be on how well the terrain features are defined by the receptor network as a whole, which is based on both the accuracy of the terrain data and the horizontal resolution of the receptor network. As noted in Section 4.3, it is advisable to check the accuracy of receptor elevations and hill height scales for significant terrain features that are likely to be associated with peak concentrations. These accuracy checks should also account for the relative elevation differences between the source and receptor since that will determine the elevation of the plume in relation to the terrain.

Given the issues and uncertainties associated with estimating the elevation at a specific location, and the potential sensitivity of AERMOD model results to differences in the relative elevations of sources and nearby receptors, users are discouraged from relying solely on AERMAP-derived source elevations in regulatory applications of AERMOD, especially for emission sources within the facility being permitted. These concerns are particularly important with newer facilities since regrading, typical with new construction, may not be reflected in the digitized terrain data. Source elevations taken from a reliable plant survey are generally considered to be the preferred option. If AERMAP-derived source elevations are used for the permitted facility, then some effort should be made to verify the accuracy of the elevations based on other reliable information, such as up-to-date topographic maps, taking into account adjustments for the horizontal datum if necessary. Use of AERMAP-derived elevations for other background sources included in the modeled inventory is generally of less concern than their use for the permitted facility, depending on the complexity of the terrain and the distances between sources within the modeled inventory. To facilitate proper review, the modeling protocol should clearly document the data and method(s) used to determine source elevations for input to AERMOD.

## 5.0 Urban applications

### 5.1 Urban/rural determination (04/2022)

The URBANOPT keyword on the CO pathway in AERMOD, coupled with the URBANSRC keyword on the SO pathway, should be used to identify sources to be modeled using the urban algorithms in AERMOD (EPA, 2026d). To account for the dispersive nature of the “convective-like” boundary layer that forms during nighttime conditions due to the urban heat island effect, AERMOD enhances the turbulence for urban nighttime conditions over that which is expected in the adjacent rural, stable boundary layer. Additionally, it defines an urban boundary layer height to account for limited mixing that may occur under these conditions. The magnitude of the urban heat island effect is driven by the urban-rural temperature difference that develops at night. AERMOD currently uses the population input on the URBANOPT keyword as a surrogate to define the magnitude of this differential heating effect. Details regarding the adjustments in AERMOD for the urban boundary layer are provided in Section 5.9 of the AERMOD model formulation document (EPA 2026a) and Cimorelli et al. (2004).

Section 7.2.1.1.b of the *Guideline* (EPA, 2024b) provides the basis for determining the urban/rural status of a source. For most applications the Land Use Procedure described in Section 7.2.1.1(b)(i) is sufficient for determining the urban/rural status. However, there may be sources located within an urban area, but located close enough to a body of water or to other non-urban land use categories to result in a predominately rural land use classification within 3 kilometers of the source following that procedure. Users are, therefore, cautioned against applying the Land Use Procedure on a source-by-source basis but should also consider the potential for urban heat island influences across the full modeling domain. Furthermore, Section 7.2.1.1(e) of Appendix W recommends modeling all sources within an **urban complex** using the urban option even if some sources may be defined as rural based on the procedures outlined in Section 7.2.1.1. Such an approach is consistent with the fact that the urban heat island is not a localized effect but is more regional in character.

Another aspect of the urban/rural determination that may require special consideration on a case-by-case basis relates to tall stacks located within or adjacent to small and up to moderate size urban areas. In such cases, the stack height, or effective plume height for very buoyant

plumes, may extend above the urban boundary layer height. The urban boundary layer height,  $z_{iuc}$ , can be calculated from the population input on the URBANOPT keyword,  $P$ , based on Equation 110 of the AERMOD formulation document (EPA, 2026a):

$$z_{iuc} = z_{iu0} \left( P/P_0 \right)^{1/4} \quad (1)$$

where  $z_{iu0}$  is the reference height of 400 meters corresponding to the reference population,  $P_0$ , of 2,000,000.

Prior to version 15181 of AERMOD, application of the urban option for these types of sources may have artificially limited the plume height resulting in anomalously high concentrations. Use of the urban option may not have been appropriate for such sources, since the actual plume was likely to be transported over the urban boundary layer and not be affected by urban enhanced dispersion. However, the potential for such anomalous results has been mitigated beginning with version 15181 of AERMOD, which has incorporated a formulation bug fix that modifies the treatment of plume rise for urban sources. Beginning with version 15181, AERMOD emulates the plume rise approach used for penetrated plumes during convective conditions if the initial plume height estimate is greater than or equal to the urban mixing height. An additional change was made in version 22112, in which the penetrated plume approach is used when the initial plume height is greater than or equal to the urban mixing height and the stack height is less than the urban mixing height. This follows more with the penetrated plume approach for convective hours and avoids dividing by zero issues. With the introduction of this formulation bug fix in version 15181 (and modification in 22112) of AERMOD, a more thorough case-specific justification will be needed, in consultation with the appropriate reviewing authority, to support excluding these elevated sources from application of the urban option.

## 5.2 Selecting population data for AERMOD's urban mode (4/2022, v22112)

For relatively isolated urban areas, the user may use published census data corresponding to the Metropolitan Statistical Area (MSA) for that location. For urban areas adjacent to or near other urban areas, or part of urban corridors, the user should attempt to identify that part of the urban area that will contribute to the urban heat island plume affecting the source(s). If this approach results in the identification of clearly defined MSAs, then census data may be used as



above to determine the appropriate population for input to AERMOD. Use of population based on the Consolidated MSA (CMSA) for applications within urban corridors is not recommended, since this may tend to overstate the urban heat island effect. Similarly, for application sites that are in isolated areas of dense population but are not representative of the larger MSA, care should be taken to determine the extent of the area the urban area that will contribute to the urban heat island plume affecting the source(s).

For situations where MSAs cannot be clearly identified, the user may determine the extent of the area, including the source(s) of interest, where the population density exceeds 750 people per square kilometer. The combined population within this identified area may then be used for input to the AERMOD model. Users should avoid using a very fine spatial resolution of population density for this purpose as this could result in significant gaps within the urban area due to parks and other unpopulated areas, making it more difficult to define the extent of the urban area. Population densities by census tract should provide adequate resolution in most cases and may still be finer resolution than desired in some cases. Since census tracts vary in size and shape, another acceptable approach would be to develop gridded estimates of population data based on census block or block group data. In such cases, a grid resolution on the order of 6 kilometers is suggested. Plotting population density with multiple “contour” levels, such as 0-500, 500-750, 750-1000, 1000-1500, etc., may also be beneficial in identifying which areas near the edge of the urban complex to include even though the population density may fall below the 750 threshold. The user should also bear in mind that the urban algorithms in AERMOD are dependent on population to the one-fourth power and are therefore not highly sensitive to variations in population. Population estimates to two significant figures should be sufficiently accurate for application of AERMOD.

### **5.3 Optional urban roughness length – URBANOPT keyword (11/2024, v24142)**

The URBANOPT keyword on the CO pathway in AERMOD (EPA, 2026d) includes an optional parameter to specify the urban surface roughness length which has a default value of 1 m when no length is specified in the input file. The urban surface roughness parameter is used to define a reference height for purposes of adjusting dispersion for surface and low-level releases to account for the enhanced turbulence associated with the nighttime urban heat island. This optional urban roughness length is not used to adjust for differences in roughness length between

the meteorological measurement site, used in processing the meteorological data, and the urban application site. Details regarding the adjustments in AERMOD for the urban boundary layer, including the use of the urban roughness length parameter, are provided in Section 5.9 of the AERMOD model formulation document (EPA, 2026a) and Cimorelli et al. (2004).

The default value of 1 meter for urban surface roughness length is considered appropriate for most applications. Any application of AERMOD that utilizes a value other than 1 meter for the urban roughness length should be based upon agreement reached between the model user and appropriate reviewing authority in accordance with Section 4.2.1(f) and pursuant to recommendations discussed in Section 7.2.1.1 of the *Guideline* (EPA, 2024b).

## **5.4 Meteorological data selections for urban applications (01/09/2008)**

### **5.4.1 Urban applications using NWS meteorological data (01/2008)**

When modeling urban sources, the urban algorithms in AERMOD are designed to enhance the turbulence levels relative to the nearby rural setting during nighttime stable conditions to account for the urban heat island effect (Cimorelli, *et al.*, 2004). For urban applications using representative NWS meteorological data, the AERMOD urban option (URBANOPT) should be selected (EPA, 2026d), regardless of whether the NWS site is in a nearby rural or an urban setting. This is because the limited surface meteorological measurements available from NWS stations will not account for the enhanced turbulence or other meteorological characteristics of the urban boundary layer included in the AERMOD urban algorithms. The determination of surface characteristics for processing NWS meteorological data for urban applications should conform to the recommendations presented in Section 3.1.

### **5.4.2 Urban applications using site-specific meteorological data (01/2008)**

In most cases, site-specific meteorological data used for urban applications should be treated in a manner similar to NWS data described in Section 5.4.1, regardless of whether the measurement site is located in a nearby rural or an urban setting. That is, the AERMOD urban option should be selected and the surface characteristics should be determined based on the recommendations in Section 3.1. This is due to the fact that the limited surface meteorological measurements available from the meteorological measurement program will not adequately account for the meteorological characteristics of the urban boundary layer included in the

AERMOD urban algorithms. However, if the measurement site is located in an urban setting and site-specific turbulence measurements are available (*e.g.*,  $\sigma_\theta$  or  $\sigma_w$ ), some adjustments to the meteorological data input to AERMOD may be necessary, as discussed in Section 3.3.

## 6.0 Source characterization

### 6.1 Capped and horizontal stacks (12/2016, v16216)

Beginning with version 16216, AERMOD includes regulatory options for modeling capped and horizontal stacks using the POINTCAP and POINTHOR source types, respectively. For capped and horizontal sources that are not subject to building downwash, the options are consistent with the approach that was approved by the Model Clearinghouse in July 1993 (see Appendix A). This approach uses an effective stack diameter to maintain the flow rate to maintain plume buoyancy, while suppressing plume momentum by setting the exit velocity to 0.001 m/s.

For capped and horizontal sources that are subject to building downwash, the options have been adapted to account for the PRIME algorithm. Since the PRIME component in AERMOD incorporates a numerical plume rise algorithm that simulates the full trajectory of the plume, AERMOD sets the initial plume trajectory angle as horizontal for the POINTHOR option. For the POINTCAP option, AERMOD assigns the initial diameter of the plume to be 2 times the actual stack diameter to account for initial spread of the plume associated with the cap. AERMOD also assigns the initial horizontal velocity of the plume for the POINTCAP option to be the initial exit velocity specified by the user divided by 4.

### 6.2 Use of area source algorithm in AERMOD (11/2024, v24142)

Plume meander for area sources was added to AERMOD version 23132 as an alpha option and retained as an alpha option in version 24142. Because of issues related to excessive run times and technical issues with model formulation, the approach has not been implemented as a regulatory option for area sources. As a result, concentration predictions for area sources may be overestimated under very light wind conditions (*i.e.*,  $u < 1.0$  m/s). The problem could arise with meteorological data derived from very low threshold instruments, such as sonic anemometers as well as data from a gridded meteorological model. Meteorological grid models can at times produce extremely light winds. During such conditions time-averaged plumes tend to spread primarily as a result of low frequency eddy translation rather than eddy diffusion. AERMOD treats this meander effect by estimating the concentration from two limiting states: 1) a coherent plume state that considers lateral diffusive turbulence only (the mean wind direction is

well defined) and 2) a random plume state (mean wind direction is poorly defined) that allows the plume to spread uniformly, about the source, in the x-y plane. The final concentration predicted by AERMOD is a weighted sum of these two bounding concentrations. Interpolation between the coherent and random plume concentrations is accomplished by assuming that the total horizontal “energy” is distributed between the wind’s mean and turbulent components.

In order to avoid overestimates for area sources during light wind conditions, it is recommended that, where possible, a volume source approximation be used to model area sources. This approach can be applied with confidence for situations in which the receptors are displaced from the source. However, for applications where receptors are located either directly adjacent to, or inside the area source, AERMOD’s area source algorithm will need to be used. For these circumstances, caution should be exercised if excessive concentrations are predicted during extremely light wind conditions. On a case-by-case basis, the reviewing authority should decide whether such low-wind predictions are unrealistic and refine modeled emissions estimates for area sources, as appropriate.

### 6.3 RLINE source type (11/2024, v24142)

The RLINE source type was first implemented as a beta option for characterizing mobile sources (i.e., roadways) in AERMOD v19191, and its use in a regulatory air modeling demonstration required alternative model approval by the Regional Office and concurrence from the Model Clearinghouse (MCH). AERMOD also limited the use of the RLINE source type to FLAT terrain only. In AERMOD version 23132, the capability to account for elevated terrain was added to the RLINE source type, and the source type was maintained as a beta option. The RLINE source type was formally promulgated as a regulatory formulation update to AERMOD beginning in version 24142 in the 2024 update to the *Guideline* (EPA, 2024b). As a regulatory option, the RLINE source type can now be used to characterize mobile sources without alternative model approval and MCH concurrence. **However, when modeling for project level transportation conformity and hot-spot analyses, refer to the EPA’s Office of Transportation and Air Quality (OTAQ) for current guidance on how to model and configure roadway sources (<https://www.epa.gov/state-and-local-transportation/project-level-conformity-and-hot-spot-analyses>).**

## 7.0 NO<sub>2</sub> Modeling (7/2026, v26135)

All Tier 2 and Tier 3 NO<sub>2</sub> screening options are coded such that NO<sub>x</sub> chemistry calculations are applied to all sources and associated combined NO<sub>x</sub> emissions in a single AERMOD model run. This ensures that the NO<sub>x</sub> chemistry schemes for each NO<sub>2</sub> screening option correctly reflect the incremental, non-linear contribution of influencing factors to conversion of NO<sub>x</sub> to NO<sub>2</sub> for each hour and at each receptor. In general, the influencing factors informing the formulation and application of each NO<sub>2</sub> screening option on a source-by-source basis include in-stack NO<sub>2</sub>/NO<sub>x</sub> ratio, NO<sub>x</sub> emissions reaction with ambient ozone, plume entrainment of ambient ozone, and dominant and ensemble plume assumptions. As such, AERMOD Tier 2 and Tier 3 NO<sub>2</sub> model outputs are a function of all sources and NO<sub>2</sub> input parameters included in the model run, and thus, the non-linear contributions from each source are mathematically tied to all sources in a single model run. This effectively means that Tier 2 and Tier 3 NO<sub>2</sub> model outputs are limited to representing one operating emissions scenario per model run. Thus, NO<sub>2</sub> model outputs by source group (e.g., PLOTFILES, MAXDCONT, etc.) are limited to being representative of contributions to all sources participating in the chemistry and total NO<sub>2</sub> modeled concentrations from all sources.

The Generic Reaction Set Method (GRSM), a Tier 3 screening option, was promulgated as a regulatory option in AERMOD beginning with version 24142 and the 2024 update to the *Guideline*. Users should note, GRSM NO<sub>2</sub> daytime concentrations are calculated differently than the other Tier 3 NO<sub>2</sub> screening options (i.e., OLM and PVMRM) given the GRSM chemistry scheme includes explicit calculations for treatment of both emitted and entrained ambient NO<sub>x</sub> (NO portion) reactions with entrained ozone and photolysis of both plume and entrained ambient NO<sub>2</sub>. GRSM daytime NO<sub>2</sub> concentration outputs by source group (e.g., PLOTFILES, MAXDCONT, etc.) depend on whether the user assigns background (as specified through the SO BACKGRND pathway) to source group ALL (i.e., SO SRCGROUP ALL BACKGROUND). If background is assigned to source group ALL, GRSM will calculate the NO<sub>x</sub>-to-NO<sub>2</sub> conversion based on the post-chemistry total NO<sub>2</sub> and NO from both source impacts and entrained ambient concentrations. If background is not assigned to source group ALL, GRSM calculates the NO<sub>x</sub>-to-NO<sub>2</sub> conversion based on the post-chemistry source NO<sub>2</sub> and NO minus a

minimum entrained ambient NO<sub>2</sub> and NO concentration (see Post-chemistry Concentrations, pg. 37, CERC, 2023). In terms of NO<sub>2</sub> NAAQS demonstrations, the former GRSM calculation would be appropriate for the cumulative impact analysis whereas the latter GRSM calculation would be more appropriate for the single source impact analysis as discussed in the *Guideline* Section 9.2.3.

GRSM nighttime NO<sub>2</sub> model concentrations are calculated similar to OLM and PVMRM, i.e., limited by ozone conversion of NO to NO<sub>2</sub> with background NO<sub>2</sub> concentrations (as specified through the SO BACKGRND pathway) simply added to the post-chemistry total NO<sub>2</sub> concentration when background is assigned to the ALL source group.

## 8.0 REFERENCES

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**Appendix A. EPA Model Clearinghouse memorandum dated July 9, 1993**

July 9, 1993

MEMORANDUM

SUBJECT: Proposal for Calculating Plume Rise for Stacks with Horizontal Releases or Rain Caps for Cookson Pigment, Newark, New Jersey

FROM: Joseph A. Tikvart, Chief  
Source Receptor Analysis Branch, TSD (MD-14)

TO: Ken Eng, Chief  
Air Compliance Branch, Region II

In response to your request, the Model Clearinghouse has reviewed your proposal for treating horizontal and capped stacks at Cookson Pigment so that the model (SCREEN or ISC2) will properly treat plume rise from the Cookson Pigment stacks. We concur in principle with the approach, with some relatively minor changes.

First, the analysis provided by New Jersey Department of Environmental Protection is technically correct. We suggest, however, that the exit velocity for horizontal and capped stacks be set to a lower figure than 0.1 m/s. A 0.1 m/s exit velocity may still result in significant momentum plume rise being calculated, even though these kinds of sources should have zero momentum rise. We therefore suggest setting the stack exit velocity to a lower value, such as .001.

For horizontal stacks that are not capped, we suggest turning stack tip downwash off, whether there are buildings or not. Stack tip downwash calculations are inappropriate for horizontal stacks.

For vertical stacks that are capped, turn stack tip downwash off and reduce the stack height by three times the actual stack diameter. The cap will probably force stack tip downwash most of the time. The maximum amount of the stack tip downwash (as calculated in ISC2) is three times the stack diameter. Reducing the stack height by this amount, while turning off the stack tip downwash option, causes the maximum stack tip downwash effect. The resulting concentrations may err slightly on the high side. For stacks with small diameters, such as those at Cookson Pigment, the error should be quite small. Note, however, that this approach may not be valid for large diameter stacks (say, several meters).

cc: A. Colecchia  
D. Wilson

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United States  
Environmental Protection  
Agency

Office of State Air Partnerships  
Air Quality Assessment Division  
Research Triangle Park, NC

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